SEPARATEDNESS APPEARS AT A FINITE STAGE UNDER AFFINE-TRANSITION LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1341-1367, following the statement at 1335-1339.

Editorial note. No omissions; the identity R_{i'} = R_i x_{U_i x U_i} (U_i x U_{i'}) is displayed.

Allowed standard prerequisites (editorial).

- Eventually closed for finite type morphisms of schemes. Limits, Lemma limits-lemma-finite-type-eventually-closed (cross-chapter, cited at ~1355-1357). R -> U_i x U is a closed immersion for some i.
- Descent of closed immersions of finite presentation. Limits, Lemma limits-lemma-descend-closed-immersion-finite-presentation (cross-chapter, cited at ~1358-1363). R_{i'} -> U_i x U_{i'} closed immersion for i' large.

Source reference. Original reference 01ZP: focused-seeds/geometry/S9-spaces-limits-descend-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0828: focused-seeds/geometry/S9-spaces-limits-descend-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084R: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1250–1250. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084R. If X is separated, then X_i is separated for some $i \in I$.*

Proof. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X_0$. For $i \geq 0$ set $U_i = U_0 \times_{X_0} X_i$ and set $U = U_0 \times_{X_0} X$. Note that U_i and U are affine schemes which come equipped with surjective étale morphisms $U_i \to X_i$ and $U \to X$. Set $R_i = U_i \times_{X_i} U_i$ and $R = U \times_X U$ with projections $s_i, t_i : R_i \to U_i$ and $s, t : R \to U$. Note that R_i and R are quasi-compact separated schemes (as the algebraic spaces X_i and X are quasi-separated). The maps $s_i : R_i \to U_i$

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and $s : R \to U$ are of finite type. By definition X_i is separated if and only if $(t_i, s_i) : R_i \to U_i \times U_i$ is a closed immersion, and since X is separated by assumption, the morphism $(t, s) : R \to U \times U$ is a closed immersion. Since $R \to U$ is of finite type, there exists an i such that the morphism $R \to U_i \times U$ is a closed immersion (Limits, Lemma 0828). Fix such an $i \in I$. Apply Limits, Lemma 01ZP to the system of morphisms $R_{i'} \to U_i \times U_{i'}$ for $i' \geq i$ (this is permissible as indeed $R_{i'} = R_i \times_{U_i \times U_i} U_i \times U_{i'}$) to see that $R_{i'} \to U_i \times U_{i'}$ is a closed immersion for i' sufficiently large. This implies immediately that $R_{i'} \to U_{i'} \times U_{i'}$ is a closed immersion finishing the proof of the lemma. $\square$